

A photograph of a hospital emergency department hallway. In the foreground, a person wearing a blue medical gown is seen from the back, slightly out of focus. The hallway leads to a set of glass doors with a red "EMERGENCY" sign above them. To the left, there is a wooden cabinet and a round clock on the wall. The lighting is bright and clinical.

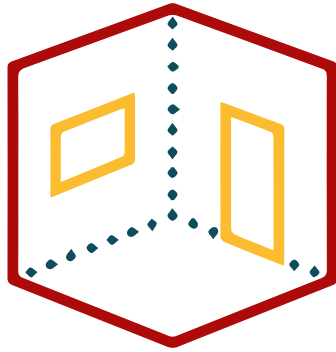
EMERGENCY

THE PEDIATRIC MBH-FRIENDLY EMERGENCY DEPARTMENT DESIGN TOOLKIT

Realizing Improved Patient Care through Human-Centered Design for Pediatric Mental and Behavioral Health
in the Emergency Department (RIPCHD.PED) Patient Safety Learning Lab



RATIONALE



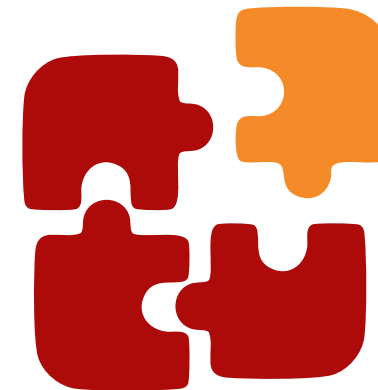
Lack of easily accessible evidence-based knowledge about the MBH-friendly ED design



No guiding structure for prioritizing MBH-friendly design and obtaining consensus from diverse stakeholders

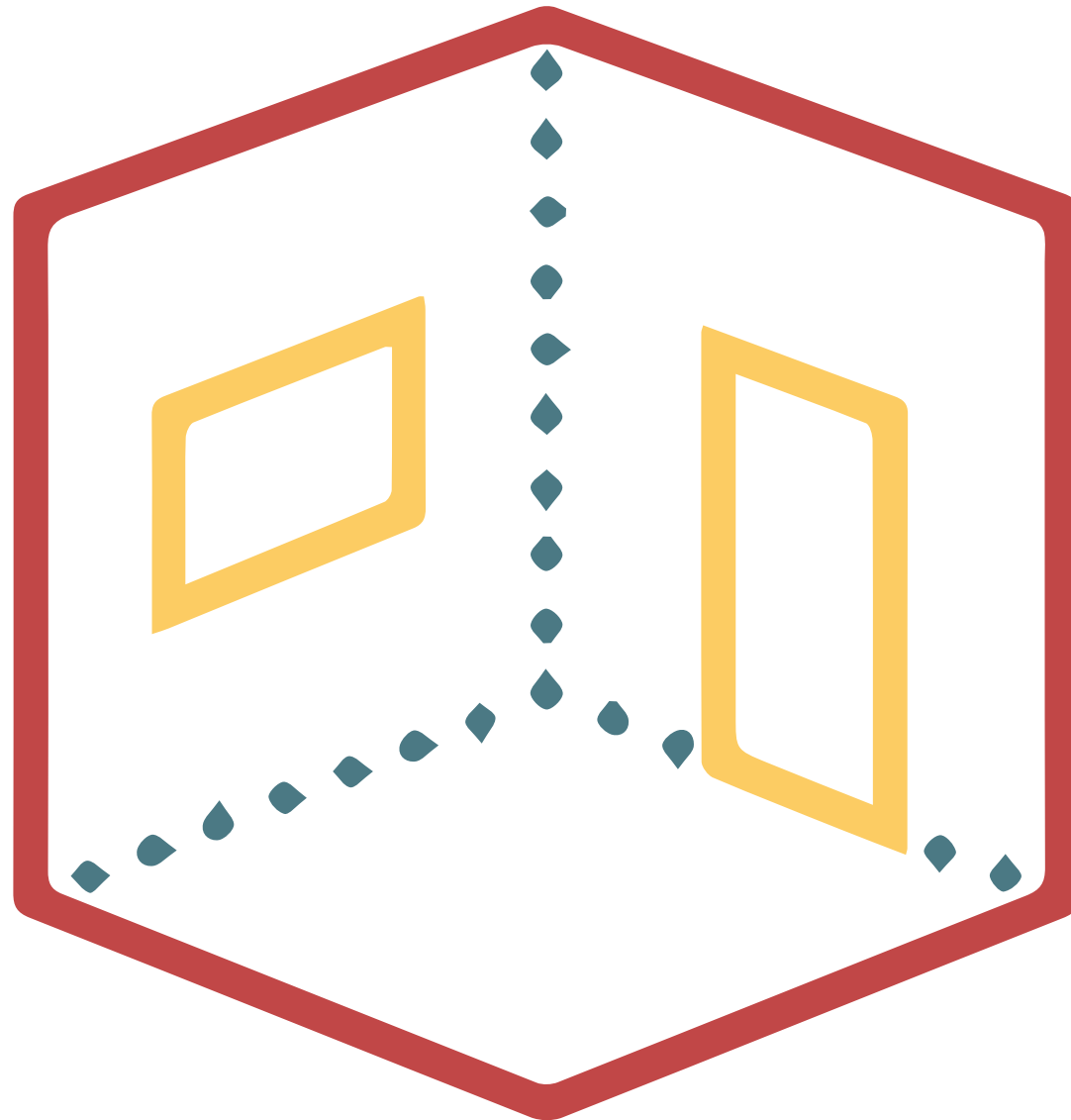


Challenges in implementing existing knowledge about MBH-friendly design in existing ED design process



Work systems perspective is missing while designing MBH-friendly EDs

PURPOSE



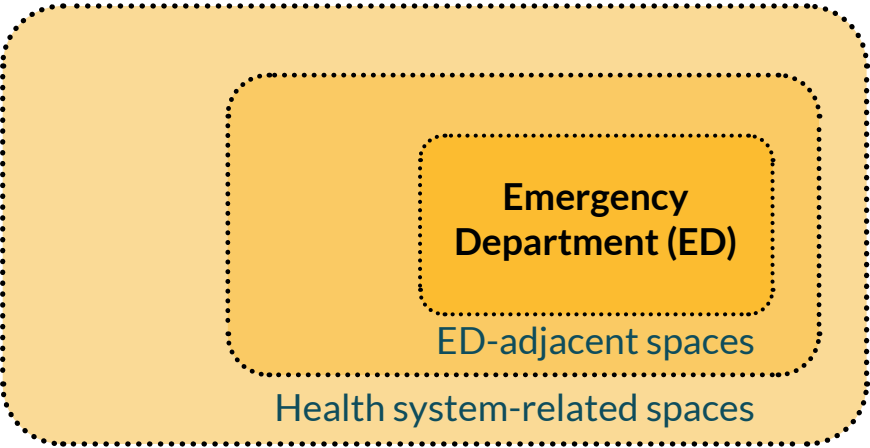
Facilitate conversation and consensus building about MBH-friendly design

Guide design teams and stakeholders to integrate MBH-friendly design considerations while designing new EDs

Ask the right MBH-care related questions during the design process.

Use existing evidence seamlessly to guide design decision-making at different points in the design process

SCOPE



Security Check



Waiting Area



Nurse Station



Patient Check-in



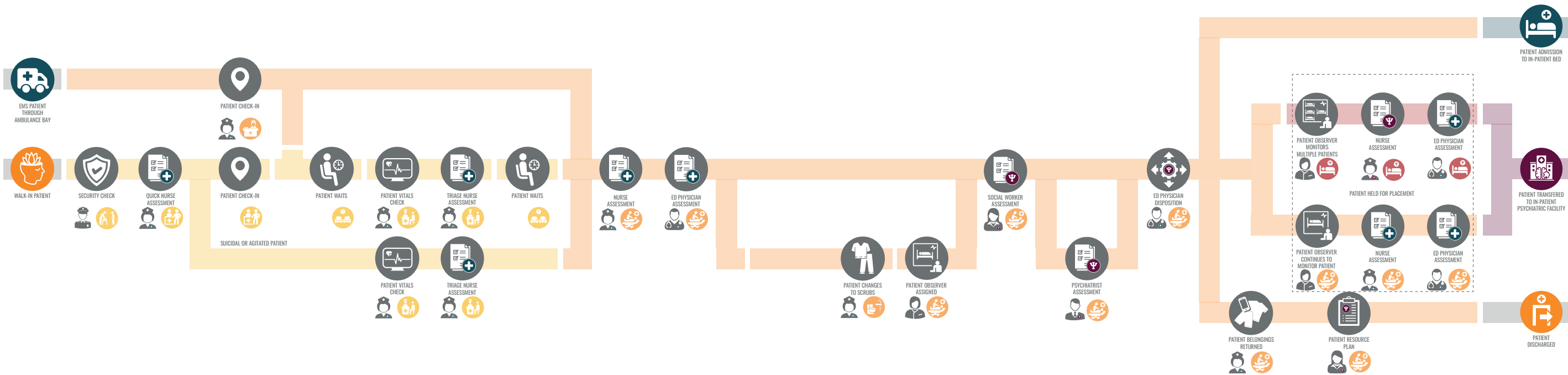
Triage Room



Patient Toilet



ED Patient Room



Pediatric MBH Patient Journey

(Soman et al., 2025)

TOOLKIT STRUCTURE

PROJECT VISION

Improve safety and experience of
children with mental and behavioral
health conditions and provider well-
being in ED

DESIGN OBJECTIVES



Improve safety & security



Optimize efficiency &
effectiveness



Improve
communication



Improve overall comfort
for all users

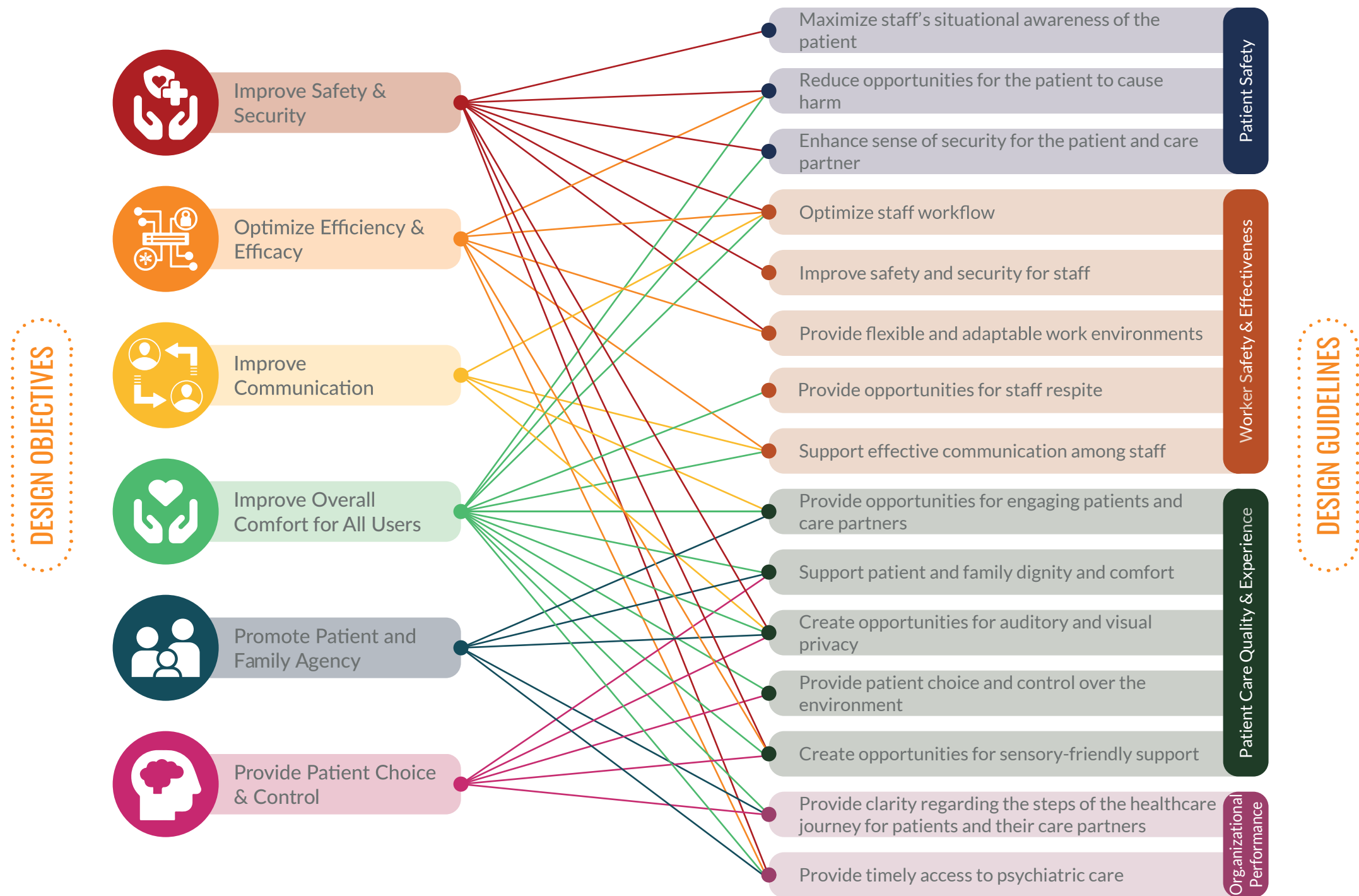


Promote patient, family,
& staff engagement

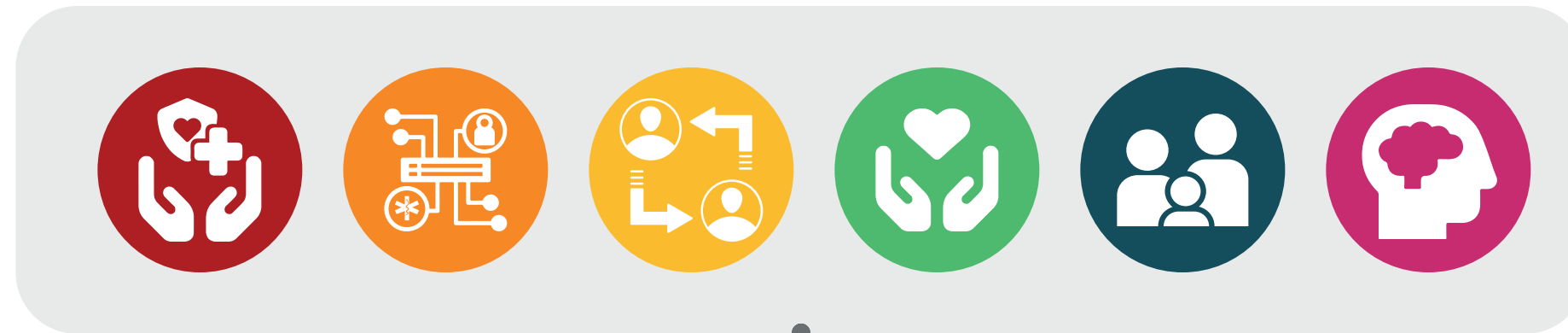


Provide patient choice
and control

TOOLKIT STRUCTURE: DESIGN GUIDELINES



TOOLKIT COMPONENTS



Pre-Planning & Programming Tool (PPT)

- ✓ Assess needs and set priorities
- ✓ Understand MBH care process and their design implications



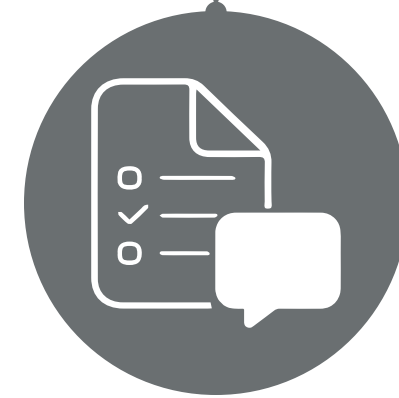
Design Alignment Tool (DAT)

- ✓ Evaluate design-goal alignment
- ✓ Inform design decision-making
- ✓ Document consensus & decisions



Mock-up Evaluation Tool (MET)

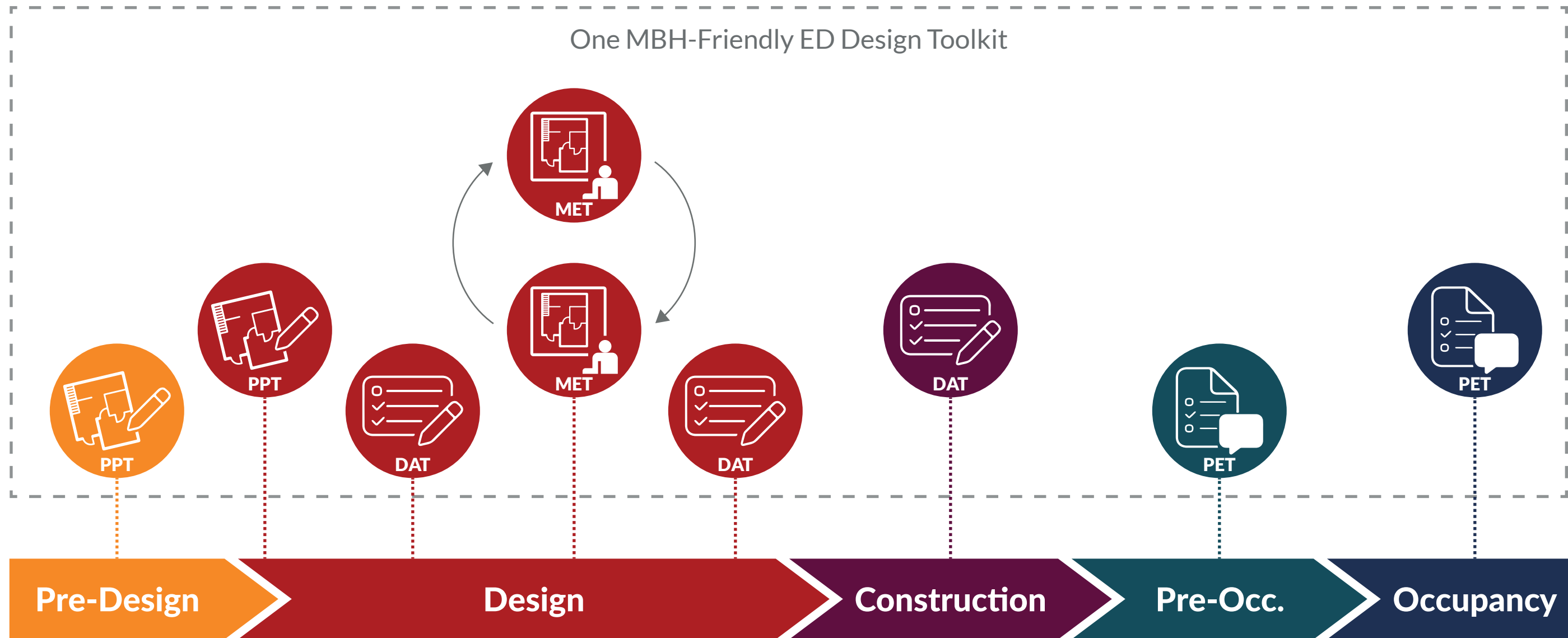
- ✓ Test design features
- ✓ Inform design decision-making
- ✓ For use with virtual reality or physical mock-ups



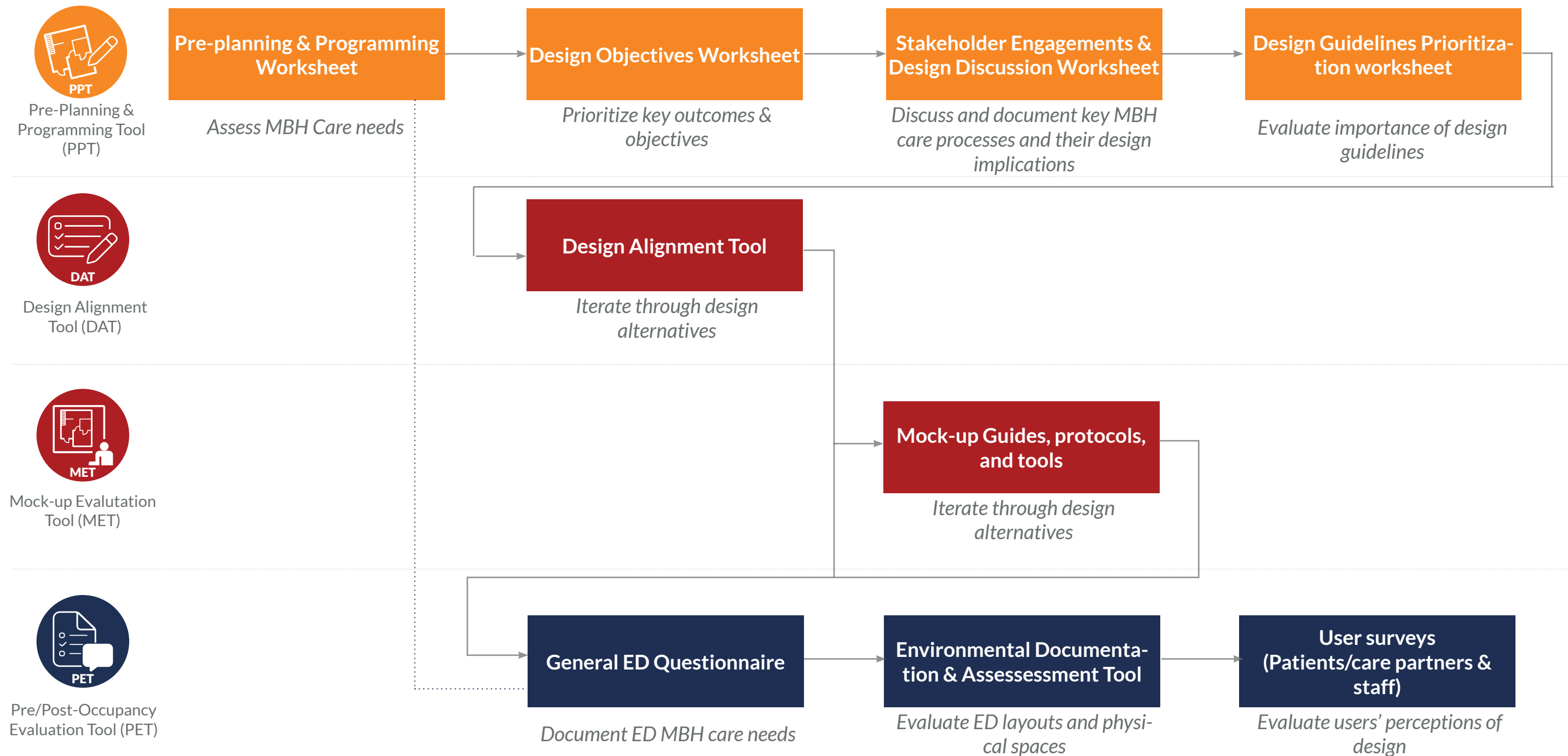
Pre/Post-Occupancy Evaluation Tool (PET)

- ✓ Evaluate the effectiveness of a design's implication in achieving project goals
- ✓ Identify areas for improvement

TOOLKIT IN THE DESIGN PROCESS

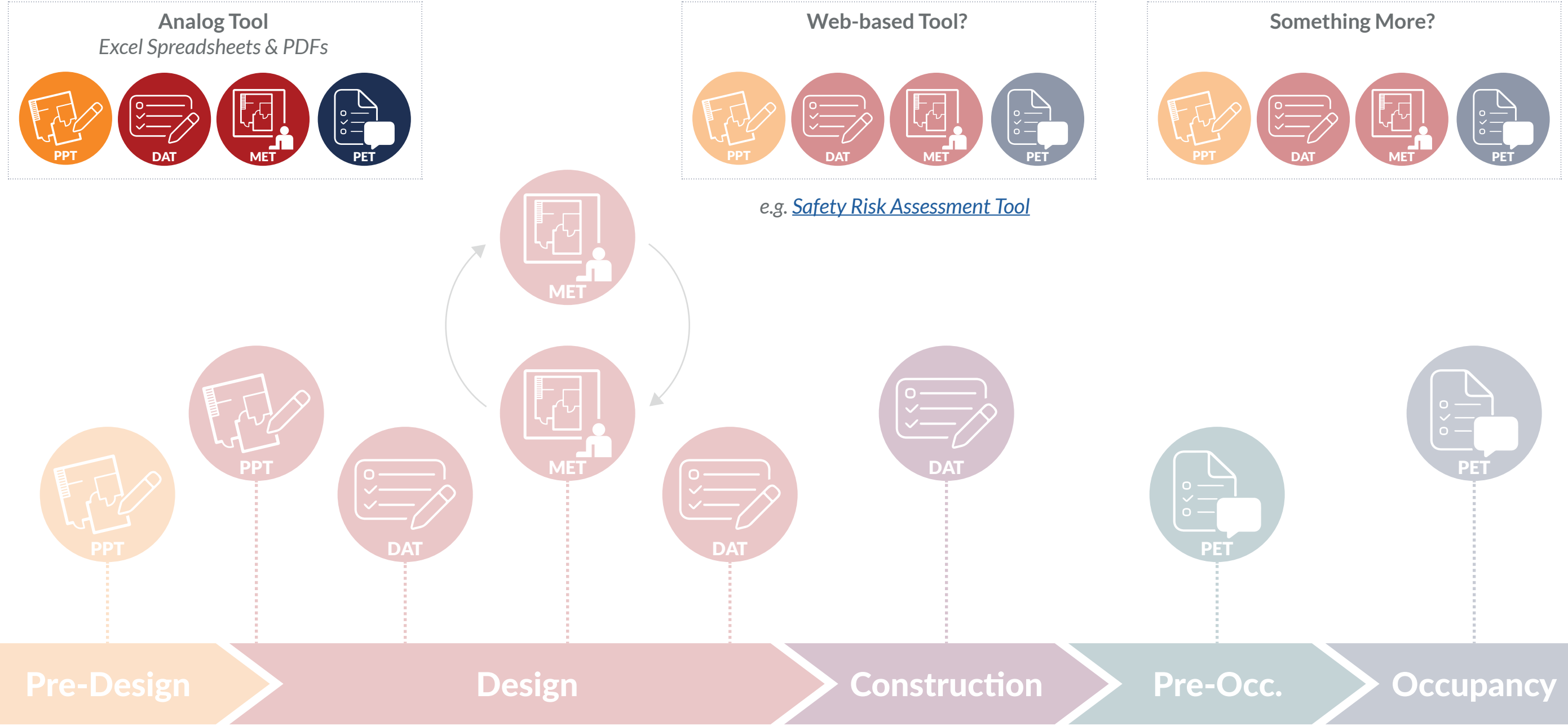


USING THE TOOLKIT



TOOLKIT DISSEMINATION

Current State → Future State ?



THANK YOU!
QUESTIONS?